

Oleksiy Nedobiychuk

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 Pisa, Italy

Education

- **M.Sc. in Computer Science (AI curriculum), University of Pisa** [2023 – Ong.] 109/110(e).
- **B.Sc. in Computer Science, University of Pisa** [2019 – 2023] 106/110.

Relevant Experience

Software Engineer at TD Group Italia S.R.L, Vecchiano, Italy [Jul. 2025 – Present]:

- Developed full-stack **Django/MySQL** platforms for agritech and healthcare IoT projects (VITE, Smart-Care, EyeDos), covering device management, role-based admin workflows, **MQTT** ingestion, and real-time updates with **Channels/Redis**.
- Built **Python** data pipelines and scheduled processors for sensor/weather data with **APScheduler**, **BeautifulSoup**, and **Requests**; improved grapevine models for phenology, maturation, irrigation, evapotranspiration, and disease-risk estimation.
- Created **Highcharts/ECharts** dashboards and data normalization tools combining **RapidFuzz**, **ONNX** embeddings, **HNSW** indexes, MySQL mapper tables, and **Keycloak**/OIDC role-based access.

Full-Stack Developer at TAI Solutions S.R.L, Pisa, Italy [Sept – Nov. 2022]:

- Built a **Spring** microservice and **Angular** application for tenant/user management, integrating **Keycloak** authentication and authorization across multiple clients.

Projects

Professional Cycling Analysis – Data Mining

- Cleaned and engineered cycling datasets, applied K-means/DBSCAN clustering, and trained a neural network to predict rider performance with 90% test accuracy.

Neural Network for Regression/Classification

- Built a neural network from scratch in **Python/NumPy** with backpropagation, Nesterov momentum, L1/L2 regularization, and grid-search tuning.

ProbLog Based AI Agent

- Built a probabilistic AI agent for a 2D fighting game with dynamic probability estimation, outperforming MCTS, MinMax, and Prolog baselines in simulation.

Web Application – Full-Stack Development

- Developed and deployed a **Svelte**/WebGL2 frontend with a **Django**/PostgreSQL backend on Debian using Nginx and Apache.

Skills

Languages	• Fluent in English, Italian, and Ukrainian.
Programming	• Proficient: Python, TypeScript/JavaScript. Familiar: C++, C, Rust. Learning: Go.
Tools	• Django, Angular, Svelte, PostgreSQL, MySQL, MongoDB, Redis, GIT, Docker.
ML/DS	• PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn.
Technologies	• Linux (Ubuntu, Arch), Shell scripting, MQTT, REST APIs, Vim, Copilot, ChatGPT.
Soft	• Problem solving, adaptability, teamwork, deep focus, goal orientation, critical thinking.
Interests	• Gym, reading, hiking, roller skating, reinforcement learning.